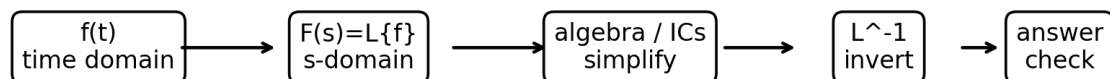


KUET MATHEMATICS - III

SECTION B CLASS NOTE

Math 2123 | Exam-oriented, source-grounded study edition



Transform -> solve algebraically -> inverse transform -> verify initial/boundary conditions

The Section B workflow: transform tools + vector differential/integral geometry.

Laplace • Inverse Laplace • Convolution • ODE/Beam • Vector Differential • Vector Integration

Credit: Made by 2223033

Prepared from the uploaded class notes, reference books and the Math 2123 question-bank analysis.

How to use this Section B note

3-out-of-4 strategy Section B has Sets B5-B8 in the analysis framework, and you answer 3. B5 is the safest anchor. Then choose two from B6, B7 and B8 according to your strength.

Set	Main block	Practical role
B5	Laplace fundamentals + direct transform	Mandatory-level: fastest high-return formula set
B6	Inverse Laplace + convolution + ODE/beam	High marks; longer algebra
B7	Vector differential calculus	Short, standardized differentiation methods
B8	Line/surface/volume integration	High-value but region setup matters

Two strong routes Route L: B5+B6+B7 (Laplace-heavy). Route V: B5+B7+B8 (Vector-heavy). Beam/ODE দুর্বল হলে Route V; multiple-integral region setup দুর্বল হলে Route L বেশি practical.

বাংলা ব্যাখ্যার নিয়ম কঠিন concept-এর বাংলা intuition দেওয়া হয়েছে; theorem/formula নাম ইংরেজিতে রাখা হয়েছে যাতে exam script-এর notation-এর সাথে মিল থাকে।

Fast navigation

Set	Must-master patterns	Memory hook
B5	table + shifts + unit step + periodic + $\sin(at)/t$	Table -> Pattern -> Property
B6	inverse method choice + convolution + ODE + beam BC	Factor -> Shift -> Convolve -> IC/BC
B7	grad/div/curl + directional derivative + surfaces + potential	G-D-C + normalize first
B8	parameterization + flux + bounds + Green/Stokes/Gauss	Curve-Surface-Volume + geometry first

VERY HIGH PRIORITY | SAFEST ANCHOR

SET B5 - Laplace Fundamentals & Direct Transform

এই set-এর মূল skill হলো pattern recognition. Function দেখে আগে ঠিক করবেন: direct table, shifting, derivative/integral property, unit step, নাকি periodic formula লাগবে।

1. Definition and existence

$$L\{f(t)\} = F(s) = \int_0^{\infty} e^{-st} f(t) dt$$

Term	Correct meaning	বাংলা cue
Piecewise continuous	On every finite interval, only finitely many finite jumps are allowed.	ভাঙা-ভাঙা হলেও finite jump
Exponential order alpha	Eventually $ f(t) \leq M e^{\alpha t}$ for some $M > 0$.	exponential-এর চেয়ে দ্রুত না বাড়া
Sufficient condition	Piecewise continuous + exponential order guarantees the transform for $s > \alpha$.	existence guarantee

Important These are sufficient conditions, not necessary conditions. Condition fail করলেই transform অবশ্যই নেই - এমন নয়; আলাদা করে convergence check লাগতে পারে.

2. Standard transform table - learn this cold

$f(t)$	$L\{f(t)\}$	Cue
1	$1/s$	constant
t^n	$n!/s^{n+1}$	power $\rightarrow$ factorial
e^{at}	$1/(s-a)$	s shifts by a
$\sin at$	$a/(s^2+a^2)$	sin $\rightarrow$ a numerator
$\cos at$	$s/(s^2+a^2)$	cos $\rightarrow$ s numerator
$\sinh at$	$a/(s^2-a^2)$	hyperbolic $\rightarrow$ minus
$\cosh at$	$s/(s^2-a^2)$	hyperbolic $\rightarrow$ minus

Mnemonic sin-এর numerator = a, cos-এর numerator = s. Circular trig-এ denominator s^2+a^2 ; hyperbolic-এ s^2-a^2 .

3. Property decision table

You see...	Use...	Formula
sum / constant multiples	Linearity	$L\{af+bg\}=aF+bG$
$e^{at} f(t)$	First shifting	$L\{e^{at} f(t)\}=F(s-a)$
$f(at), a>0$	Scaling	$L\{f(at)\}=(1/a)F(s/a)$
$t^n f(t)$	Differentiate $F(s)$	$L\{t^n f\}=(-1)^n d^n F/ds^n$
$f'(t), f''(t), \dots$	Derivative transform	$L\{f'\}=sF-f(0)$; extend recursively
integral from 0 to t	Integral property	$L\{\int_0^t f(u)du\}=F(s)/s$
$f(t)/t$	Division by t	$L\{f(t)/t\}=\int_s^{\infty} F(u)du$ (when valid)

You see...	Use...	Formula
repeat every T	Periodic formula	$[\int_0^T e^{(-st)} f(t) dt] / [1 - e^{(-sT)}]$
switch at $t=a$	Unit step / 2nd shift	$u(t-a), e^{(-as)}$ factor

Exam shortcut একটি direct-transform problem-এ integration শুরু করার আগে 10 seconds property scan করুন। Table/property দিয়ে হলে direct definition integral করা সময় নষ্ট করে।

4. Derivatives - initial conditions enter automatically

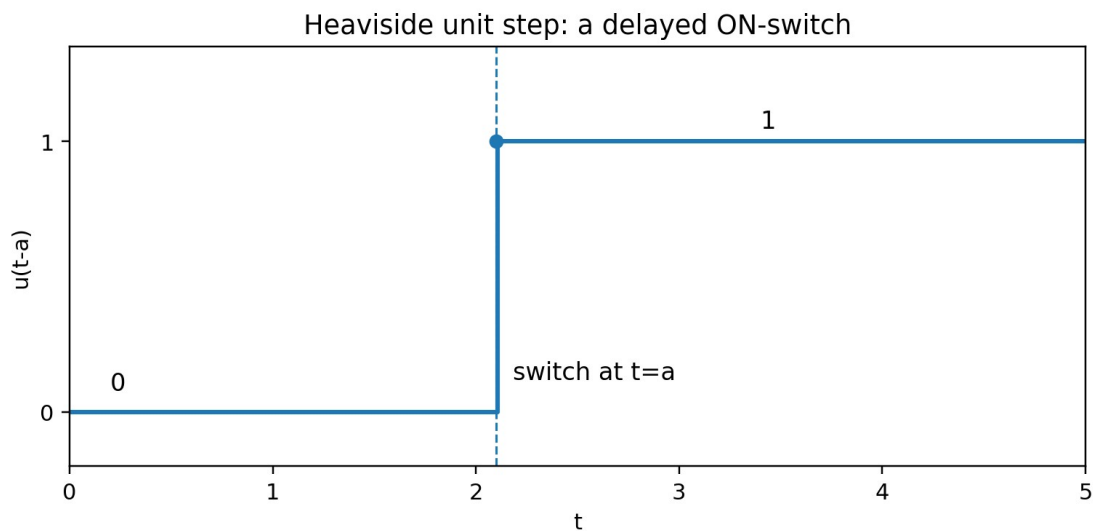
$$L\{f'(t)\} = sF(s) - f(0)$$

$$L\{f''(t)\} = s^2 F(s) - s f(0) - f'(0)$$

$$L\{f^{(n)}(t)\} = s^n F(s) - s^{(n-1)} f(0) - \dots - f^{(n-1)}(0)$$

Memory hook Derivative order যত বাড়ে, s-এর power এক ধাপ করে নামতে থাকে এবং initial data এক ধাপ করে যোগ হয়.

5. Unit step / Heaviside function



$u(t-a)$ acts like an ON-switch at $t=a$.

$$u(t-a)=0 \text{ for } t < a, \quad u(t-a)=1 \text{ for } t \geq a \quad \text{and} \quad L\{u(t-a)\} = e^{(-as)}/s$$

$$L\{u(t-a) f(t-a)\} = e^{(-as)} F(s)$$

$$L\{g(t)u(t-a)\} = e^{(-as)} L\{g(t+a)\}$$

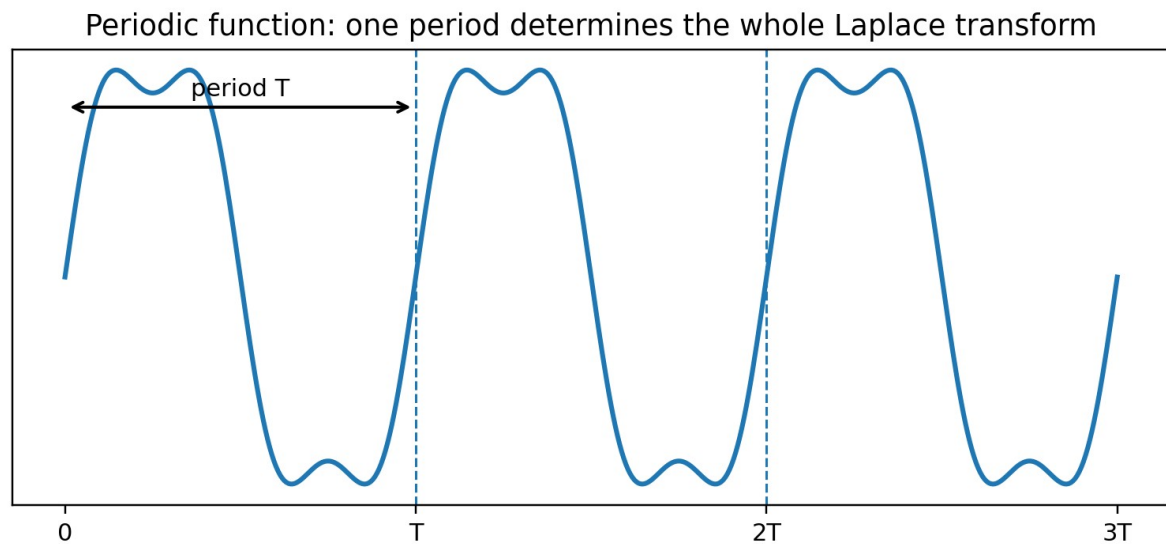
general form when the factor is not already written as $f(t-a)$

Piecewise -> step trick Base expression লিখুন, তারপর প্রতিটি switching point-এ “new value - old value” jump-কে $u(t-a)$ দিয়ে যোগ করুন। এটিই সবচেয়ে safe construction rule.

Source-matched example: 2019 Q3(c) family

Question-bank compilation explicitly includes “Define unit step function and express a piecewise function in unit-step form.” তাই definition + step conversion একই সাথে practice করুন।

6. Periodic functions



Only one period 0 to T is integrated; the denominator accounts for all repetitions.

$$\text{If } f(t+T)=f(t), \text{ then } L\{f(t)\} = \left[\int_0^T e^{-st} f(t) dt \right] / [1 - e^{-sT}]$$

1. Identify the smallest convenient period T .
2. Integrate only over one full period.
3. Keep the denominator $1 - e^{-sT}$.
4. If the first period is piecewise, split only inside 0 to T .

Common mistake Periodic function-এর transform-এ denominator $1 - e^{-sT}$ বাদ গেলে answer incomplete.

7. High-priority existence pair: $\sin(at)/t$ vs $\cos(at)/t$

$$L\{\sin(at)/t\} = \int_s^\infty a/(u^2 + a^2) du = \tan^{-1}(a/s) \quad (a > 0, s > 0)$$

For $\cos(at)/t$, as $t \rightarrow 0^+$, $e^{-st}\cos(at)/t$ behaves like $1/t$. Therefore the defining improper integral diverges near zero, so its Laplace transform does not exist.

Direct repeat This “ $\cos(at)/t$ does not exist” item appears as 2015 Q7(c) and 2024 Q3(c)(ii) in the uploaded theory/proof compilation.

8. Source-matched direct example

Lecture example: $f(t)=t/k$ for $0 < t < k$ and $f(t)=1$ for $t > k$. Split the defining integral at $t=k$:

$$L\{f(t)\} = \int_0^k (t/k)e^{-st}dt + \int_k^\infty e^{-st}dt = [1-e^{-ks}]/(ks^2)$$

Technique Piecewise function হলে breakpoint-এ integral split করুন. Integration by parts-এ LIATE cue (Log, Inverse trig, Algebraic, Trig, Exponential) class note-এ ব্যবহার করা হয়েছে.

Matching questions from the question bank - B5

Year / Q	Matching question family	Priority
2019 Q3(b)	Laplace transform definition	★★★
2020 Q4(a)	Laplace definition / core transform theory	★★★
2022 Q5(a)	Laplace transform definition	★★★
2018 Q5(a)	Sufficient conditions for existence	★★★
2019 Q3(c)	Unit step definition + piecewise representation	★★★
2020 Q4(a)	Piecewise/unit-step related Laplace family	★★
2021 Q5(c)	Delayed sine / unit-step family	★★★
2015 Q7(c)	Why $L\{\cos(at)/t\}$ does not exist	★★★
2024 Q3(c)(ii)	Same $\cos(at)/t$ existence question repeated	★★★
2019/21/22/24	Periodic or piecewise transform families recur	★★★

B5 rapid self-test

Prompt	Target response
$e^{3t} \sin 2t$	shift $s \rightarrow s-3$ in $2/(s^2+4)$
$t \cos at$	$-d/ds [s/(s^2+a^2)]$
$f(t)$ repeats every T	periodic formula with denominator
piecewise switch at a	unit step + e^{-as}
f'' with initial data	$s^2F - sf(0) - f'(0)$

VERY HIGH PRIORITY | LONG BUT STRUCTURED

SET B6 - Inverse Laplace, Convolution & ODE / Beam

এই set-এ সবচেয়ে বড় advantage হলো method selection. Denominator দেখে inverse method বেছে নিলে algebra অনেক কমে যায়। ODE/beam-এ transform নেওয়ার সঙ্গে সঙ্গে IC/BC লিখুন।

1. Inverse Laplace - method selector

Pattern in F(s)	Best first method	Recognition clue
factorable rational function	Partial fractions	linear/repeated/quadratic factors
quadratic in shifted s	Complete square + 1st shift	$(s-a)^2+b^2$
simple distinct denominator roots	Heaviside expansion	fast residues $P(a_k)/Q'(a_k)$
product of two known transforms	Convolution	$F(s)G(s)$
$e^{(-as)}F(s)$	Second shifting / unit step	time delay a

Decision rule Factor first. তারপর প্রশ্ন করুন: “shift আছে?”, “product আছে?”, “simple roots আছে?” Method নিজে থেকেই বের হবে.

2. Partial fractions and completing the square

Illustrative partial-fraction pattern

$$(2s+5)/[(s+1)(s+2)] = 3/(s+1) - 1/(s+2)$$

$$\mathcal{L}^{-1} = 3e^{-t} - e^{-2t}$$

Shifted quadratic patterns

$$\mathcal{L}^{-1}\{(s+2)/[(s+2)^2+9]\} = e^{-2t} \cos 3t$$

$$\mathcal{L}^{-1}\{1/[(s+2)^2+9]\} = (1/3)e^{-2t} \sin 3t$$

Complete-square cue $s^2+4s+13 = (s+2)^2+9$. Numerator-এ s থাকলে $(s+2)-2$ করে split করুন.

3. Heaviside expansion formula

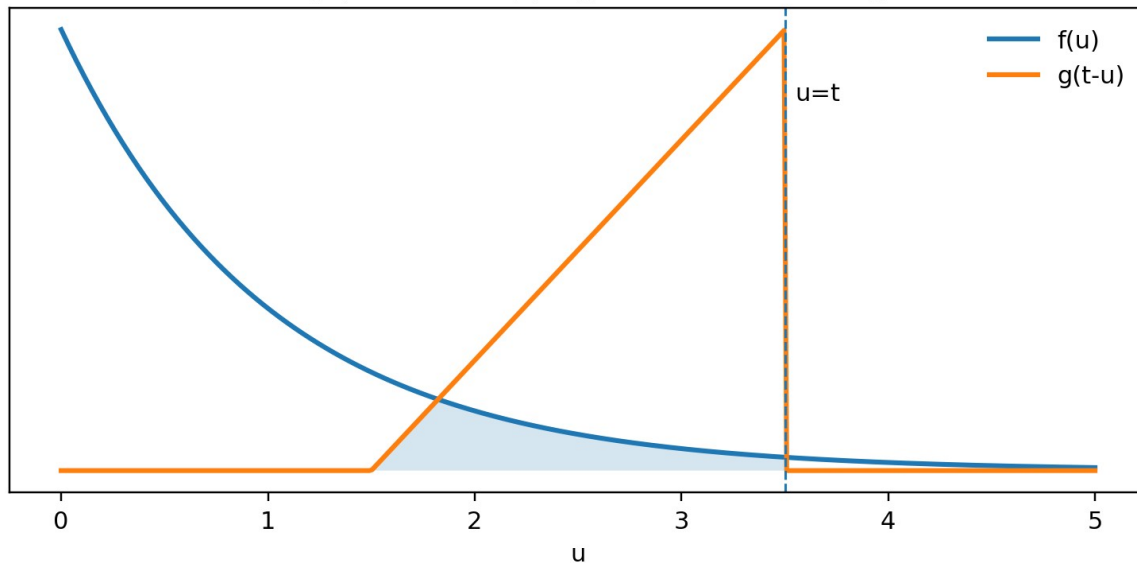
If $Q(s)$ has distinct simple roots a_k : $P(s)/Q(s) = \sum [P(a_k)/Q'(a_k)] * 1/(s-a_k)$

$$\mathcal{L}^{-1}\{P/Q\} = \sum [P(a_k)/Q'(a_k)] e^{(a_k t)}$$

Restriction এই short formula simple distinct roots-এর জন্য। Repeated roots হলে ordinary partial fraction / repeated-pole formula ব্যবহার করুন.

4. Convolution theorem

Convolution: multiply the overlapping functions and integrate from 0 to t



Convolution combines two known inverse transforms over $0 \leq u \leq t$.

$$\mathcal{L}^{-1}\{F(s)G(s)\} = (f*g)(t) = \int_0^t f(u) g(t-u) du$$

বাংলা **intuition** দুটি signal/function-কে একে অপরের উপর slide করিয়ে overlap-এর product যোগ করা = convolution.

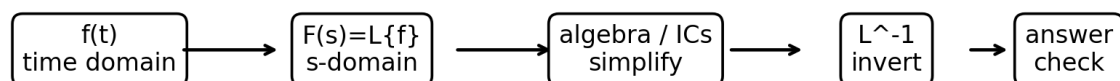
Source worked example

For $1/[s(s^2+a^2)]$, use $f(t)=1$ and $g(t)=\sin(at)/a$:

$$\mathcal{L}^{-1}\{1/[s(s^2+a^2)]\} = \int_0^t \sin(a(t-u))/a du = [1-\cos(at)]/a^2$$

When convolution wins Partial fraction awkward কিন্তু product clearly matches table transforms হলে convolution often cleaner.

5. ODE by Laplace - repeatable workflow



Transform -> solve algebraically -> inverse transform -> verify initial/boundary conditions

For IVPs, the transform changes differential equations into algebraic equations in $Y(s)$.

1. Let $Y(s)=\mathcal{L}\{y(t)\}$.
2. Transform every derivative and insert initial conditions immediately.

3. Collect $Y(s)$ terms and solve algebraically.
4. Use partial fractions/shifting/convolution to invert.
5. Check $y(0)$, $y'(0)$, etc. from the final answer.

Source-aligned IVP example

$$y'' - 4y' + 4y = 64 \sin 2t, \quad y(0)=0, y'(0)=1$$

$$Y(s) = [1+128/(s^2+4)]/(s-2)^2$$

after applying the derivative transforms and ICs

$$y(t) = (17t-8)e^{(2t)} + 8 \cos 2t$$

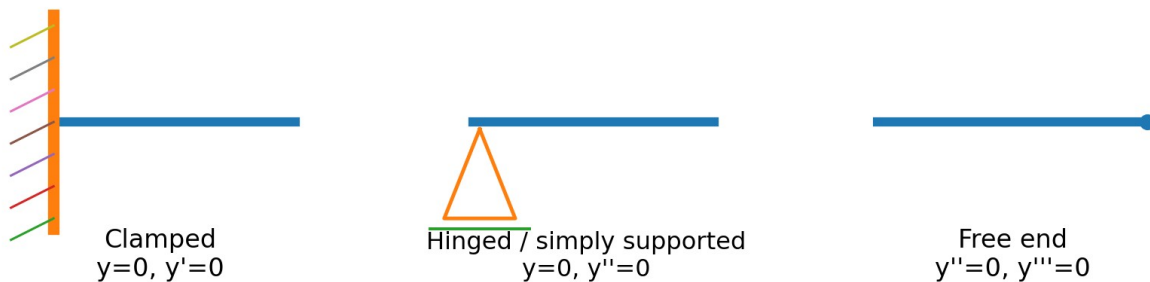
Check At $t=0$: $y=0$ and $y'=1$. IC check is a fast way to catch sign/algebra errors.

6. Beam deflection by Laplace

$$EI y''''(x) = w(x)$$

E = Young's modulus; I = area moment of inertia; $w(x)$ = load per unit length

Beam boundary-condition memory sketch



These boundary conditions are explicitly used in the uploaded beam compilation.

End type	Boundary conditions	Physical meaning
Clamped / embedded	$y=0, y'=0$	no deflection, no slope
Hinged / simply supported	$y=0, y''=0$	no deflection, no bending moment
Free	$y''=0, y'''=0$	no bending moment, no shear force

Boundary mnemonic Clamp locks y and slope. Hinge locks y but allows rotation, so moment $y''=0$. Free end has neither moment nor shear: $y''=y'''=0$.

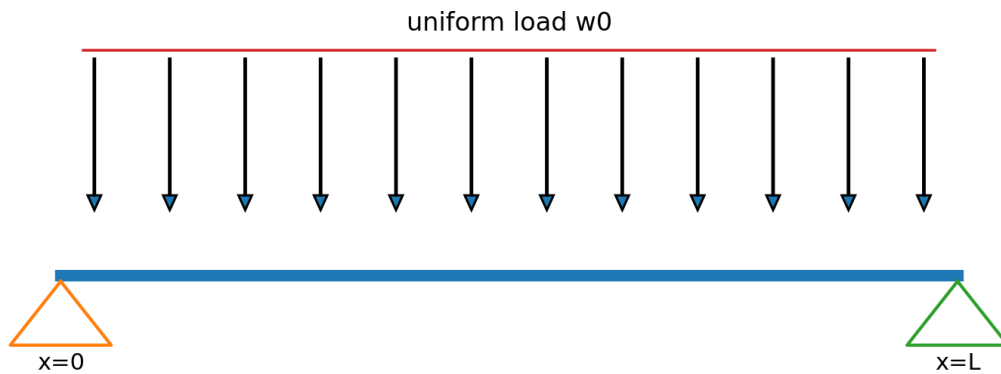
Discontinuous load -> unit step

$$\text{UDL } w_0 \text{ acting only for } 0 < x < a: \quad w(x) = w_0[1 - u(x-a)]$$

Then the transform automatically produces $e^{(-as)}$ terms; after inverse transform they become shifted powers multiplied by $u(x-a)$.

Continuous UDL, simply supported beam

Simply supported beam under continuous UDL

*Reference geometry for the 2022 simply-supported continuous-UDL problem.*

$$y(0)=0, y''(0)=0, y(L)=0, y''(L)=0$$

$$y(x) = [w_0/(24EI)] [x^4 - 2Lx^3 + L^3 x]$$

for the sign convention $EI y^{(4)} = +w_0$; reverse sign if downward load is defined negative**7. Beam question map from the uploaded compilation**

Year / Q	Beam type	Load
2017 Q2(b)	Cantilever: clamped-free	Discontinuous UDL, $0 < x < L/2$
2018 Q6(b)	Clamped-clamped	Discontinuous triangular load
2019 Q8(a)	Clamped-simply supported	Discontinuous UDL, $0 < x < L/2$
2022 Q7(a)	Simply supported / hinged	Continuous UDL

Exam tactic Beam-এ আগে sketch + four BC লিখুন, তারপর load $w(x)$. Boundary conditions ভুল হলে পুরো long solution drift করে.

Matching questions from the question bank - B6

Year / Q	Matching question family	Priority
2015-2025	Inverse Laplace of rational functions: near-annual family	★★★
2015/18/19	Convolution-based inverse families	★★★
2021-2025	Convolution variants continue in recent papers	★★★
2016/17/19	ODE solved by Laplace	★★★
2020-2025	IVP/forcing ODE variants recur	★★★
2017 Q2(b)	Cantilever beam by Laplace	★★★
2018 Q6(b)	Clamped beam with triangular/discontinuous load	★★
2019 Q8(a)	Mixed clamped-supported beam	★★★
2022 Q7(a)	Simply supported continuous UDL	★★★

B6 rapid self-test

If you see...	First thought
$e^{-3s}F(s)$	delay 3 $\rightarrow$ unit step after inverse
$(s-2)^2+9$	first shift by +2 in time exponential e^{2t}
product $F(s)G(s)$	consider convolution
4th-order beam equation	write four BC before algebra
simple distinct denominator roots	Heaviside expansion may be fastest

B6 last-minute error shield

Risk	Safe habit
Partial fraction sign slips	Substitute one easy s -value back into the decomposition to check.
Shift sign confusion	$F(s-a) \leftrightarrow e^{at}f(t)$; $F(s+a) \leftrightarrow e^{-at}f(t)$.
ODE initial data lost	Write derivative transform with ICs before simplifying.
Convolution variable confusion	Use $f(u)g(t-u)$, integrate u from 0 to t .
Beam constants wrong	Apply all four BC only after the inverse transform/4 integrations are in hand.

Method marks Even if the algebra becomes long, clearly write $Y(s)$, the chosen inverse method, and all IC/BC. These lines protect substantial method marks.

30-second inverse sanity checks

- If $F(s) \sim 1/s$ as $s \rightarrow \infty$, $f(0+)$ is often finite/nonzero; if $F(s)$ decays faster, $f(0+)$ may be zero (initial-value theorem only when its hypotheses apply).
- An e^{-as} factor must create a time delay and a unit-step factor in the inverse.
- For real rational $F(s)$, complex quadratic factors should combine into real sin/cos terms.

SET B7 - Vector Differential Calculus

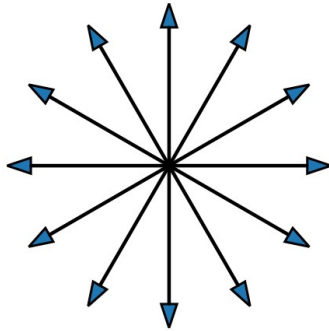
এই set-এ প্রায় সব problem-এর শুরু nabla operator দিয়ে। মূল habit: point substitute করার আগে symbolic grad/div/curl বের করুন; directional derivative-এ direction vector অবশ্যই unit করুন।

1. Nabla master table

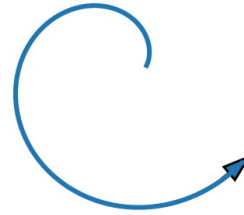
$$\nabla = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}$$

Operation	Formula	Input -> output	Meaning
Gradient	$\phi = (\phi_x, \phi_y, \phi_z)$	scalar -> vector	fastest increase / normal
Divergence	$\nabla \cdot \mathbf{F} = P_x + Q_y + R_z$	vector -> scalar	source/sink strength
Curl	$\nabla \times \mathbf{F}$	vector -> vector	local rotation
Laplacian	$\nabla^2 \phi = \phi_{xx} + \phi_{yy} + \phi_{zz}$	scalar -> scalar	div(grad ϕ)

Positive divergence
source / outward flux



Nonzero curl
local spinning / circulation

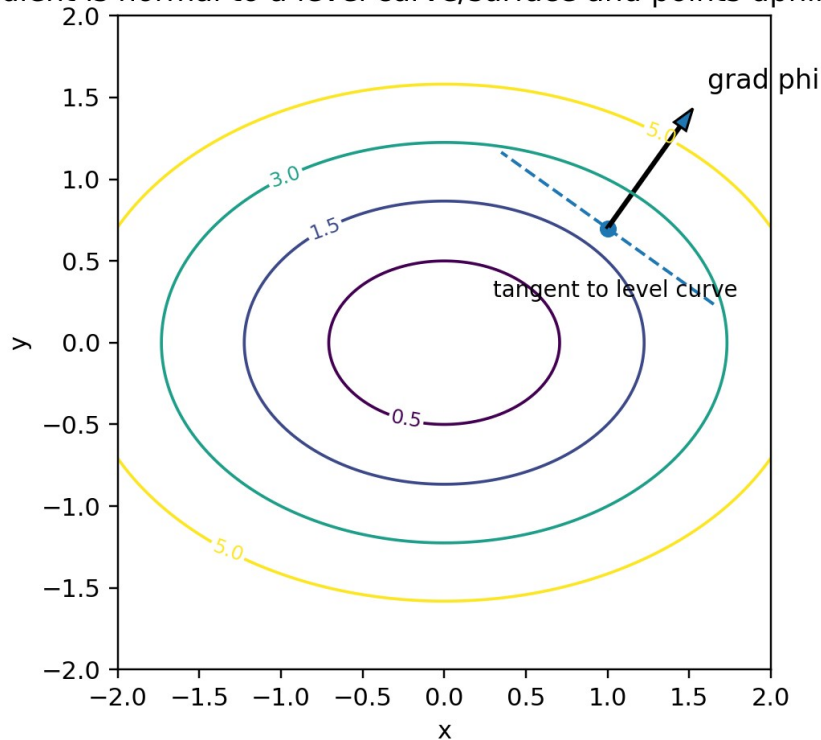


Divergence measures net spreading; curl measures local rotational tendency.

Physical picture grad = পাহাড়ে সবচেয়ে খাড়া uphill arrow; div = ছোট volume থেকে net বের হওয়া/টোকা; curl = ছোট paddle wheel ঘোরার tendency.

2. Gradient and directional derivative

Gradient is normal to a level curve/surface and points uphill fastest



grad phi is perpendicular to phi=constant and gives the maximum rate direction.

Directional derivative along unit vector $\hat{u}$: $D_{\hat{u}} \phi = \nabla \phi \cdot \hat{u}$

Maximum directional derivative = $|\nabla \phi|$, direction = $\nabla \phi / |\nabla \phi|$

Normalize first Direction যদি $d=(a,b,c)$ দেওয়া থাকে, $\hat{u}=d/|d|$ না করে dot product করলে directional derivative ভুল scale হবে.

Source example

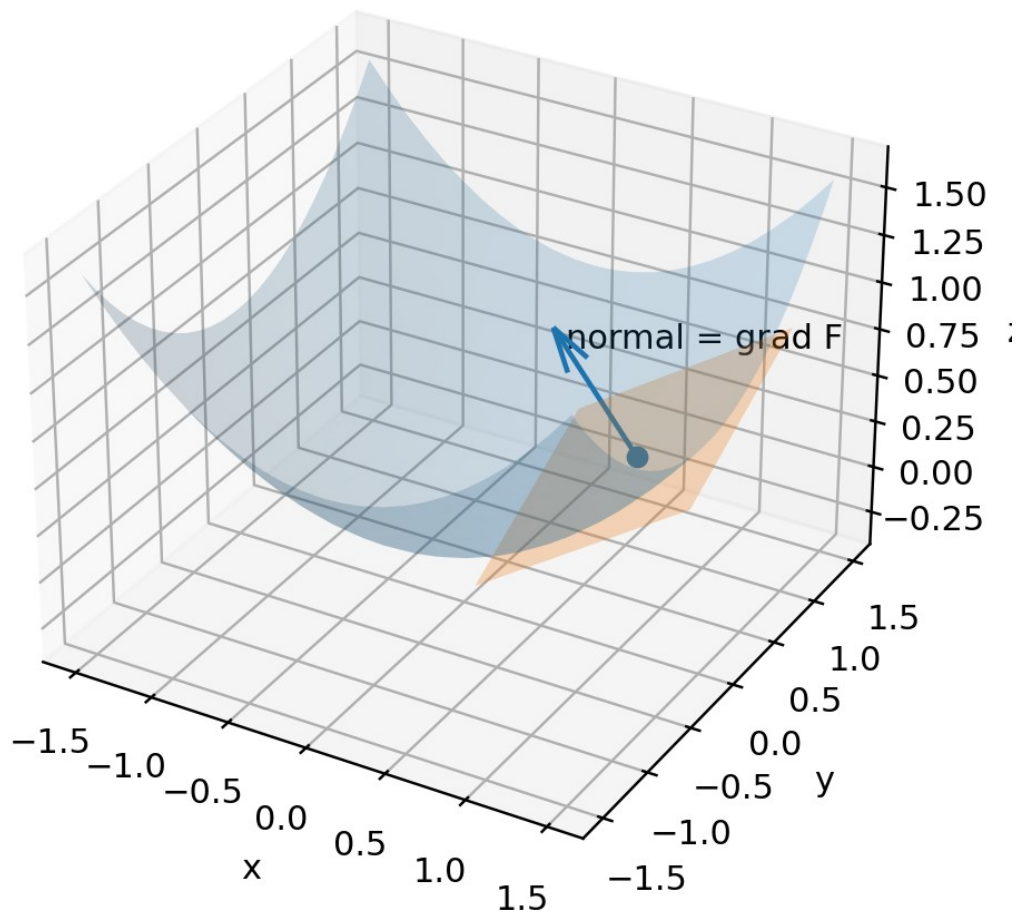
$\phi=x^2-y^2+2z^2$ at $P(1,2,3)$, in direction from P to $Q(5,0,4)$.

$$\nabla \phi(P)=(2,-4,12), \quad PQ=(4,-2,1), \quad \hat{u}=(4,-2,1)/\sqrt{21}$$

$$D_{\hat{u}} \phi = (2,-4,12) \cdot (4,-2,1)/\sqrt{21} = 28/\sqrt{21}$$

3. Level surface, tangent plane and normal line

Tangent plane and normal from the gradient



For $F(x,y,z)=0$, the gradient at P is normal to the tangent plane.

$$\text{For } F(x,y,z)=0 \text{ at } P(x_0,y_0,z_0): \nabla F(P) \cdot [(x,y,z)-P] = 0$$

$$\text{Normal line: } (x-x_0)/F_x(P) = (y-y_0)/F_y(P) = (z-z_0)/F_z(P)$$

বাংলা cue Surface-এর tangent direction-এ F বদলায় না, তাই $dF = \nabla F \cdot dr = 0$. অর্থাৎ ∇F tangent-এর perpendicular - এটিই normal.

4. Angle between two surfaces

$$\cos \theta = |\nabla F \cdot \nabla G| / (|\nabla F| |\nabla G|) \quad \text{at the common point}$$

$$\text{Surfaces orthogonal} \Leftrightarrow \nabla F \cdot \nabla G = 0$$

Technique Surface equation আগে $F=0$, $G=0$ form-এ আনুন; gradients symbolicভাবে বের করে intersection point বসান.

5. Solenoidal, irrotational and conservative

Word	Test	বাংলা meaning / consequence
Solenoidal	$\text{div } \mathbf{F} = 0$	net source/sink নেই
Irrotational	$\text{curl } \mathbf{F} = 0$	local rotation নেই
Conservative	$\mathbf{F} = \text{grad } \phi$	work path-independent; potential exists

Correct theorem condition On a simply connected domain, with suitable continuous partial derivatives, $\nabla \times \mathbf{F} = 0$ implies $\mathbf{F}$ is conservative. Domain condition বাদ দিলে holes/singularities-এর জন্য exception হতে পারে.

Potential-finding workflow

1. From $\phi_x = P$ integrate P with respect to x : $\phi = \int P \, dx + h(y, z)$.
2. Differentiate that ϕ with respect to y and compare with Q to find h_y .
3. Integrate h_y in y , leaving a function of z .
4. Use $\phi_z = R$ to determine the remaining function.
5. Add an arbitrary constant C .

Shortcut Potential বের করার সময় “missing variables” function $h(y, z)$, then $g(z)$ রাখতে ভুলবেন না.

6. Physical meanings of grad, div and curl

Quantity	Physical meaning	Exam phrase
$\text{grad } \phi$	Direction of greatest increase; magnitude is max rate; normal to level surface.	gradient = steepest ascent
$\text{div } \mathbf{F}$	Net outward flux per unit volume; positive source, negative sink.	divergence = source strength
$\text{curl } \mathbf{F}$	Axis/magnitude of local rotational tendency; for rigid rotation $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$, $\text{curl } \mathbf{v} = 2\boldsymbol{\omega}$.	curl = local spin

Exact historical theory The uploaded theory compilation lists physical meaning of Gradient and Divergence explicitly as 2015 Q5(c).

7. High-value vector identities

Identity	Memory cue
$\text{curl}(\text{grad } \phi) = 0$	$\text{curl grad} = 0$
$\text{div}(\text{curl } \mathbf{F}) = 0$	$\text{div curl} = 0$
$\text{div}(\phi \mathbf{F}) = \text{grad } \phi \cdot \mathbf{F} + \phi \text{div } \mathbf{F}$	product rule: div
$\text{curl}(\phi \mathbf{F}) = \text{grad } \phi \times \mathbf{F} + \phi \text{curl } \mathbf{F}$	product rule: curl
$\text{Laplacian } \phi = \text{div}(\text{grad } \phi)$	$\text{Laplacian} = \text{div grad}$

Proof technique Component form-এ mixed partial derivatives equal (e.g., $\phi_{xy} = \phi_{yx}$) ব্যবহার করলে $\text{curl grad} = 0$ এবং $\text{div curl} = 0$ দ্রুত prove হয়.

Matching questions from the question bank - B7

Year / Q	Matching question family	Priority
2015 Q5(c)	Physical meaning of gradient and divergence	★★★
2016/17/19	Directional derivative / maximum-rate family	★★★
2020/22/23	Directional derivative variants continue	★★★
2015/16/18	Angle between / orthogonal surfaces	★★★
2021/22/23	Surface-angle / normal geometry variants	★★★
2015,17-19	Solenoidal / irrotational / potential family	★★★
2021-23	Conservative-field/potential variants	★★★
2016/17/21-22	Tangent plane / normal line family	★★★
Multiple years	Vector identities involving grad/div/curl	★★

B7 rapid self-test

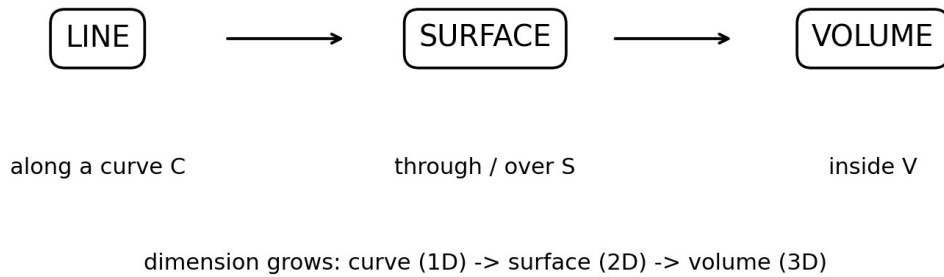
Question word	Immediate formula
maximum rate	$ \text{grad } \phi $
direction of max rate	$\text{grad } \phi / \text{grad } \phi $
orthogonal surfaces	$\text{grad } F \cdot \text{grad } G = 0$
solenoidal	$\text{div } F = 0$
irrotational	$\text{curl } F = 0$
normal to $F=0$	$\text{grad } F$

VERY HIGH PRIORITY | GEOMETRY FIRST

SET B8 - Vector Integration: Line, Surface & Volume

এই set-এর arithmetic-এর চেয়ে setup বেশি গুরুত্বপূর্ণ। Sketch $\rightarrow$ parameter/bounds $\rightarrow$ integrand - এই order বজায় রাখলে ভুল কমে।

1. Integral-type master map



Think dimension first: curve, surface or volume?

Integral	Typical form	Physical/geometric meaning
Line / work	$\text{integral_C } \mathbf{F} \cdot d\mathbf{r}$	work / circulation along a path
Scalar line	$\text{integral_C } f \, ds$	scalar accumulation along curve length
Surface flux	$\text{double integral_S } \mathbf{F} \cdot \mathbf{n} \, dS$	flow through a surface
Volume	$\text{triple integral_V } f \, dV$	total quantity inside a 3D region

2. Line integral / work done - universal algorithm

$$\mathbf{W} = \int_C \mathbf{F} \cdot d\mathbf{r} = \int [P \, dx + Q \, dy + R \, dz]$$

1. Parameterize the curve: $\mathbf{r}=\mathbf{r}(t)$, $a \leq t \leq b$.
2. Compute $d\mathbf{r}=\mathbf{r}'(t)dt$.
3. Substitute $\mathbf{F}(\mathbf{r}(t))$.
4. Take the dot product $\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)$.
5. Integrate with the correct orientation.

Straight line shortcut From A to B, use $\mathbf{r}(t)=\mathbf{A}+t(\mathbf{B}-\mathbf{A})$, $0 \leq t \leq 1$. Piecewise path হলে each segment আলাদা parameterize করুন.

Repeated 2016 Q7(b) & 2017 Q4(c)

For $\mathbf{A}=(2y+3)\mathbf{i} + xz\mathbf{j} + (yz-x)\mathbf{k}$ from O to (2,1,1):

Path	Setup	Result
Three straight segments	$O \rightarrow (0,0,1) \rightarrow (0,1,1) \rightarrow (2,1,1)$	10
One straight segment	$x=2t, y=t, z=t, 0 \leq t \leq 1$	8

Interpretation Same endpoints but different results means this field/path pair is path-dependent. Conservative হলে endpoints fixed থাকলে work একই হতো.

3. Green theorem - closed planar curve

$$\oint_C P dx + Q dy = \iint_R (\partial Q / \partial x - \partial P / \partial y) dA$$

C positively oriented (counter-clockwise)

When to use Closed curve in xy-plane + $Pdx+Qdy$ দেখলে Green theorem first choice. Orientation clockwise হলে sign flips.

Question-bank examples

2018 Q8(b) and 2019 Q6(b) both use triangular closed paths; 2023 Q2(a) asks circulation around a circle and reduces immediately to area via Green theorem.

4. Surface flux

$$\text{Flux} = \iint_S \mathbf{F} \cdot \mathbf{n} dS$$

Surface description	Useful vector area element
$z=g(x,y)$, upward	$\mathbf{n} dS = (-g_x, -g_y, 1) dx dy$
$x=g(y,z)$, +x direction	$\mathbf{n} dS = (1, -g_y, -g_z) dy dz$
implicit $G(x,y,z)=0$	normal direction is $\text{grad } G$; choose orientation carefully

Unit normal vs vector area If you use unit $\hat{n}$, multiply by dS . If you use vector area element $\mathbf{n} dS$ directly, do not multiply by another dS factor.

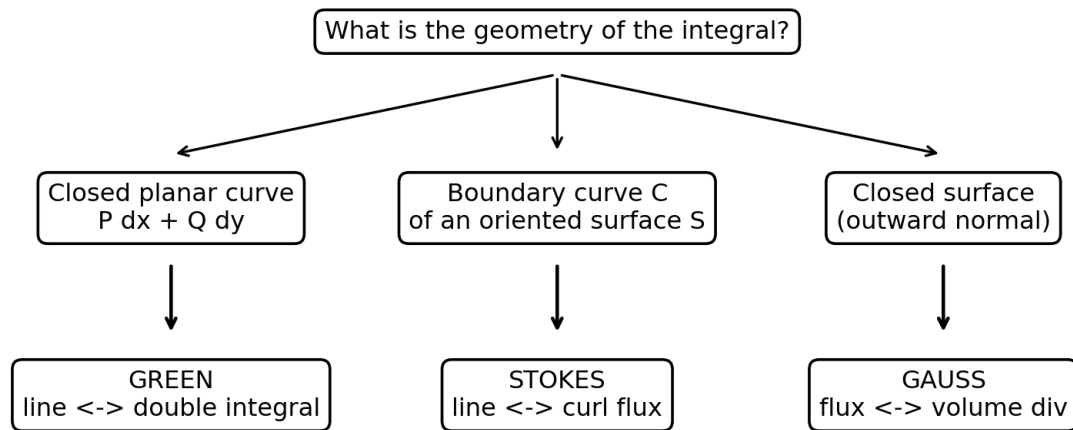
2015 Q7(a) source example

$\mathbf{F} = yz \mathbf{i} + zx \mathbf{j} + xy \mathbf{k}$ over the unit sphere in the first octant. On $x^2+y^2+z^2=1$, outward $\mathbf{n}=(x,y,z)$, so $\mathbf{F} \cdot \mathbf{n} = 3xyz$. With spherical coordinates:

$$x = \sin \phi \cos \theta, y = \sin \phi \sin \theta, z = \cos \phi, dS = \sin \phi d\phi d\theta$$

$$0 \leq \phi \leq \pi/2, 0 \leq \theta \leq \pi/2 \quad \Rightarrow \quad \text{flux} = 3/8$$

5. Stokes and Gauss - theorem selector



Shortcut: planar closed boundary -> Green | surface boundary -> Stokes | closed surface -> Gauss

Select the theorem from geometry before integrating.

$$\text{Stokes: } \oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, dS$$

C and n obey the right-hand orientation rule

$$\text{Gauss: } \iint_{\partial V} \mathbf{F} \cdot \mathbf{n} \, dS = \iiint_V (\nabla \cdot \mathbf{F}) \, dV$$

surface must be closed; n is outward

Mnemonic Green = flat/planar closed boundary; Stokes = boundary curve of a surface; Gauss = closed surface enclosing volume.

6. Volume integral and coordinate systems

System	Coordinates	Volume element
Cartesian	x, y, z	$dV = dx \, dy \, dz$
Cylindrical	$x = r \cos \theta, y = r \sin \theta, z = z$	$dV = r \, dr \, d\theta \, dz$
Spherical	$x = \rho \sin \phi \cos \theta, y = \rho \sin \phi \sin \theta, z = \rho \cos \phi$	$dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$

Jacobian memory Cylindrical adds r. Spherical adds $\rho^2 \sin(\phi)$. এই factors বাদ পড়া সবচেয়ে common marks-loss.

Region setup rule

1. Sketch the solid / projection first.
2. Choose coordinates that match symmetry: cylinder -> cylindrical; sphere/cone -> spherical often helps.
3. Write inner-variable bounds from surfaces.
4. Write outer bounds from the projection.
5. Only then substitute the integrand and Jacobian.

7. Closed-surface source example: 2021 Q3(a)

The compiled question uses a quarter cylinder $x^2 + y^2 \leq 9$ in the first quadrant, $0 \leq z \leq 2$. For $\mathbf{q} = 4yx \mathbf{i} - 2y^2 \mathbf{j} - z^2 \mathbf{k}$:

$$\text{div } \mathbf{q} = 4y - 4y - 2z = -2z$$

$$\text{Cylindrical bounds: } 0 \leq r \leq 3, 0 \leq \theta \leq \pi/2, 0 \leq z \leq 2, \quad dV = r \, dz \, dr \, d\theta$$

$$\text{Flux} = \iiint (-2z) \, r \, dz \, dr \, d\theta = -9\pi$$

Why Gauss is easier Closed cylinder-এর individual curved/flat surface flux আলাদা না করে one triple integral of divergence যথেষ্ট.

8. Exact question map from the vector-integration compilation

Area	Year / Q examples	Pattern
Line / work	2015 Q6(c); 2016 Q7(b); 2017 Q4(c)	spiral + piecewise/straight paths
Green / circulation	2018 Q8(b); 2019 Q6(b); 2023 Q2(a)	triangle/circle closed path
Work variants	2020 Q2(b); 2021 Q2(c); 2022 Q2(c); 2024 Q2(a)(ii)	curved/parametric path
Surface flux	2015 Q7(a); 2018 Q8(c); 2019 Q6(c); 2021 Q3(a)	sphere/cylinder/plane/closed surface
More surface flux	2022 Q3(a); 2023 Q2(b); 2024 Q2(b)	recurring flux geometry
Volume integrals	2016 Q7(c); 2019 Q7(a); 2021 Q3(b); 2022 Q3(c); 2023 Q3(a); 2024 Q2(c)	bounded 3D regions

Matching questions from the question bank - B8

Year / Q	Matching question family	Priority
2015 Q6(c)	Line integral on a spiral	★★
2016 Q7(b) / 2017 Q4(c)	Same field: piecewise path vs direct straight path	★★★
2018 Q8(b)	Green theorem on triangle	★★★
2019 Q6(b)	Green theorem triangle variant	★★★
2020 Q2(b) / 2024 Q2(a)(ii)	Work along constrained curve	★★★
2021 Q3(a)	Closed-surface flux; Gauss theorem	★★★
2015 Q7(a)	Sphere first-octant surface flux	★★★
2018 Q8(c) / 2023 Q2(b)	Cylinder surface-flux family	★★★
2016/19/21-24	Volume-integral families	★★★

B8 rapid self-test

Geometry clue	Tool / setup
closed planar C	Green
C is boundary of surface S	Stokes
closed surface	Gauss
straight segment A→B	$\mathbf{r} = \mathbf{A} + t(\mathbf{B} - \mathbf{A})$
sphere / cone symmetry	consider spherical

LAST-REVISION PAGES

Section B Formula Sheet & Final Strategy

1. One-page Laplace essentials

Topic	Formula
Definition	$L\{f\} = \int_0^{\infty} e^{-st} f(t) dt$
Powers	$L\{t^n\} = n! / s^{n+1}$
Trig	$L\{\sin at\} = a / (s^2 + a^2)$; $L\{\cos at\} = s / (s^2 + a^2)$
First shift	$L\{e^{at} f\} = F(s-a)$
Second shift	$L\{u(t-a) f(t-a)\} = e^{-as} F(s)$
Derivative	$L\{f'\} = s^2 F - sf(0) - f'(0)$
Integral	$L\{\int_0^t f\} = F/s$
Multiply t^n	$L\{t^n f\} = (-1)^n F^{(n)}(s)$
Periodic	$L\{f\} = [\int_0^T e^{-st} f(t) dt] / [1 - e^{-sT}]$
Convolution	$L^{-1}\{FG\} = \int_0^t f(u) g(t-u) du$

2. One-page vector essentials

Topic	Formula
Gradient	$\text{grad } \phi = (\phi_x, \phi_y, \phi_z)$
Directional derivative	$D_u \phi = \text{grad } \phi \cdot \hat{u}$; $\max = \text{grad } \phi $
Divergence	$\text{div } F = P_x + Q_y + R_z$
Curl	$\text{curl } F = \text{grad} \times F$
Tangent plane	$\text{grad } F(P) \cdot (r - P) = 0$
Angle surfaces	$\cos \theta = \text{grad } F \cdot \text{grad } G / (\text{grad } F \text{grad } G)$
Line work	$\int_C F \cdot dr$
Green	$\text{closed } \int_C (P dx + Q dy) = \text{double } \int_R (Q_x - P_y) dA$
Stokes	$\text{closed } \int_C F \cdot dr = \text{double } \int_S \text{curl } F \cdot n dS$
Gauss	$\text{double } \int_{\text{closed } S} F \cdot n dS = \text{triple } \int_V \text{div } F dV$

3. Proof / theory priority list

Priority	Theory / proof	Why
A	Laplace definition + sufficient existence conditions	direct short-question family
A	Linearity / first shifting / derivative-transform proofs	core lecture properties
A	Unit step definition + second shifting usage	recurs with piecewise functions
A	Why $L\{\cos(at)/t\}$ does not exist	direct 2015/2024 repeat
A	Convolution theorem statement + application	high-frequency inverse method
A	Physical meaning of gradient/divergence/curl	short theory + intuition
A	$\text{curl grad} = 0$, $\text{div curl} = 0$	common vector-identity proof
A	Green / Stokes / Gauss statements + orientation conditions	integration theorem selection

4. Common mistakes to eliminate

Mistake	Correct habit
Using a non-unit direction in directional derivative	normalize the direction first
Forgetting $1 - e^{-sT}$ in periodic transform	write theorem before integration
Wrong sign in first shift	$e^{at}f(t) \rightarrow F(s-a)$
Treating $\text{curl}=0$ as conservative without domain condition	mention simply connected domain
Mixing unit normal with vector area element	choose one approach only
Using Gauss on an open surface	Gauss requires closed surface; close it or use direct flux
Wrong Stokes orientation	right-hand rule links boundary direction and normal
Missing Jacobian r or $\rho^2 \sin \phi$	write dV before bounds
Beam BC memorized incorrectly	use clamp y, y' ; hinge y, y'' ; free y'', y'''

5. Final 3-set decision

Situation	Recommended 3 sets	Reason
Balanced / safer default	B5 + B7 + strongest of B6/B8	B5 and B7 are short, reusable anchors
Strong in Laplace algebra	B5 + B6 + B7	avoids long 3D integration setup
Strong in vector geometry/integration	B5 + B7 + B8	avoids beam/ODE algebra
B6 beam looks unfamiliar in exam	B5 + B7 + B8	switch to vector route
B8 region/bounds look ugly	B5 + B6 + B7	switch to Laplace route

Final rule Prepare all four at minimum formula level, but make three sets “exam-ready”. In the hall, scan all B5-B8 first; choose based on the exact subquestions, not only the set label.

6. Source basis used for this note

- Math 2123 Question Bank Analysis (Made by 2223033) - B5-B8 topic distribution and repetition priorities.
- Laplace Lec - 02/03/04/05 - standard table, direct piecewise method, shifting, derivatives, integrals and periodic formula.
- Theory and Problems of Laplace Transforms - Murray R. Spiegel - existence, inverse methods, convolution and ODE applications.
- kuet-math3-unit-step-functions.pdf - 2015-2024 unit-step / piecewise exam compilation.
- kuet-math3-beam-deflections.pdf - governing equation, boundary conditions and 2017/2018/2019/2022 beam questions.
- kuet-math3-theory-and-proofs.pdf - Laplace theory plus gradient/divergence theory and vector proof families.
- 1.-H.K.-Dass-Advanced-Engineering-Mathematics.pdf - vector differential calculus and transform reference formulas.
- kuet-math3-vector-integrations.pdf - 2015-2024 line, surface and volume integration questions and verified worked solutions.

Accuracy note Where a formula depends on orientation or sign convention (surface normal, Stokes boundary direction, beam load sign), the convention is stated explicitly. This prevents “correct method, wrong sign” errors.